DAVID YAVO

Forward Deployed AI Engineer | Applied ML | Production Software Engineering
581-777-7562 • davidaureleyavo@gmail.com • Québec (QC) relocating to Toronto, Canada
github • LinkedIn

PROFESSIONAL SUMMARY

Applied ML engineer with a **research master's in privacy-preserving synthetic data** and **8+ years of production software engineering** spanning real-time data pipelines, GPU rendering, and cross-layer performance diagnosis. Designed and open-sourced a model-agnostic generation pipeline evaluated across three real-world datasets, and built an open-sourced an LLM agent on the Anthropic SDK with native tool use that automates a multi-step evaluation methodology end to end. Bilingual French/English. Targeting applied ML, ML software engineering, and forward-deployed engineering roles.

APPLIED ML RESEARCH & PROJECTS

LLM evaluation agent *agentic workflow on the Anthropic SDK, tabular-synth-eval-agent, open source, 2026*

- Built an autonomous evaluation agent on the **Anthropic SDK with native tool use**: five typed Python tools (Jensen-Shannon distance, Cohen's d, TSTR/TRTR utility gap via XGBoost, HEOM nearest-neighbour identifiability for privacy, and TF-IDF retrieval over metric documentation so no threshold is written into the prompt)
- Built an evaluation harness for the agent itself: eight golden real/synthetic pairs with answers known by construction, mechanical checks of tool selection, numeric faithfulness (every number in a report traceable to a tool result) and verdict correctness, plus an LLM judge for clarity; measured 100% tool selection and verdict accuracy and 99.8–100% numeric faithfulness across three Claude models over 54 runs, with the noise floor stated
- Built a regression suite that replays saved agent mistakes through the checkers in seconds, with no model calls, so a fixed failure stays fixed; 46 unit tests cover the tools and API. Linting, type checks and all tests run in GitHub Actions on every pull request
- Exposed the agent as a FastAPI web service with input validation, a concurrency limit and per-run logs of latency, tokens and cost; containerized it (multi-stage, non-root Docker image) and set up GitHub Actions to build and publish the image on every merge, with an Azure Container Apps deployment prepared but not yet live
- Added an LLM tabular-generation baseline (schema and marginal statistics only, strict JSON structured output, per-row validation) and benchmarked it against CTGAN and TVAE on Adult, Cardio and Credit with the same four tools and the same agent

Privacy-Preserving Synthetic Data Pipeline *master's thesis + journal paper*

- Designed a **model-agnostic post-processing pipeline** for deep generative models (CTGAN, TVAE) producing synthetic tabular data
- Implemented a layer-frozen **mode-patching algorithm** restoring missing categorical support, and **HEOM-kNN privacy filtering** reducing proximity-based disclosure risk
- Evaluated across three real-world datasets (Credit, Adult, Cardio) on 8+ fidelity, utility, and privacy metrics
- Reduced categorical divergence by up to **36%** and improved dependence preservation 10-14%, while keeping predictive utility within **~1%** of the unprocessed baseline

TECHNICAL SKILLS

Machine learning: PyTorch, deep generative models (CTGAN, TVAE), XGBoost, scikit-learn, privacy metrics (DCR, HEOM identifiability, attribute inference, Anonymeter), distribution comparison and effect sizes

LLM applications & agentic workflows: Anthropic SDK, native tool use, agent orchestration, structured outputs, prompt design, retrieval over documentation (TF-IDF index), LLM-as-judge

AI evaluation: evaluation harness design, golden cases, numeric-faithfulness and tool-sequence checks, regression suites for agents, model evaluation, run logging (latency, tokens, cost)

AI-assisted development: Claude Code, GitHub Copilot

Data & Statistics: Real-time data pipelines, HDF5 (h5py), Mixed-type preprocessing and encoding, distribution comparison, effect sizes, privacy metrics

Programming: Python, C#/.NET, C++, SQL, HLSL

Engineering practice & delivery: Git/GitLab, reviewed pull requests, pytest, ruff, mypy, GitLab CI, Jenkins, GitHub Actions, Docker (multi-stage, non-root), FastAPI, architecture and specification documents, code review

Systems & Tools: DirectX 11, GPU profiling, Git/GitLab, Linux, Django, REST/SOAP services, unit testing, code review

PROFESSIONAL EXPERIENCE

Software Developer Evident (formerly Olympus NDT, now Wabtec)

Québec, Canada | 2020 – Present

Phased array ultrasonic testing instruments (OmniScan line): embedded acquisition software and PC-side analysis software for industrial non-destructive inspection.

- Led discovery for a new inspection view requested by vendors to match competitor devices: worked with the system engineer, product owner and director to pin down what they actually needed, the performance constraints and the MVP scope; shipped a CPU-computed MVP plus two GPU proofs of concept (VTK and Direct3D 11), giving leadership real performance numbers to decide the feature and which rendering technology to reuse for future work
- Design and implement **GPU visualization features in C#, DirectX 11 and HLSL shaders**, rendering ultrasonic amplitude data as inspection views; owned a feature rendering amplitude along the true beam path inside a part, including meshing, sample windowing, and single-pass max-amplitude compositing in the pixel shader
- Architect features **end to end across presentation, business, and data layers**, pushing cursor-driven selection down to a data-layer compression gate with targeted file replay
- Optimize **real-time pipeline performance**: replaced full-file replay with targeted band replay over multi-gigabyte **HDF5** inspection files, and resolved UI-blocking threading bottlenecks
- Diagnose defects spanning physics, data, and GPU layers, **proving root cause by measurement** rather than inspection: GPU readback, HDF5 file analysis in Python, and arithmetic verification of replayed data volumes
- Perform in-depth code review for a 15-developer Scrum team, author architecture and specification documents, and validate fixes on physical instruments

Software Developer / Programmer Analyst R3D

Québec, Canada | 2019 – 2020

- Contributed to the stabilization and evolution of enterprise systems (ECO-D platform) for government and public-sector clients
- Designed and implemented tools to manage operational workflows and data processing
- Collaborated directly with stakeholders to translate requirements into robust, maintainable solutions

Programmer Analyst Référence Système (Optel Vision)

Québec, Canada | 2017 – 2019

- Developed **Python REST/SOAP integration middleware** connecting industrial software to third-party enterprise systems
- Implemented backend components and configured web services using the **Django** framework
- Developed testing strategies including unit tests and validation tooling
- Contributed to code reviews and collaborative development processes

EDUCATION

Master of Computer Science (Research) Université Laval, Québec, Canada

2023 – 2026 | Thesis: Generating High-Quality Privacy-Preserving Synthetic Data (submitted, under evaluation)
Graduate Faculty Honour Roll (Tableau d'honneur, Faculté des études supérieures et postdoctorales), 2026

Master of Computer Science - Artificial Intelligence Université Laval, Québec, Canada

2021 – 2023 | Transitioned to the research-based program in 2023

Bachelor of Mathematics & Computer Science Université Laval, Québec, Canada

2014 – 2017

University Technical Diploma in Networks and Telecommunications Institut national polytechnique Houphouët Boigny, Côte d'Ivoire

2009 – 2012

LANGUAGES

French: Native **English:** Professional working proficiency